

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	$n = \frac{pV}{RT} = \frac{1.2 \times 10^5 \times 1.6 \times 10^{-3}}{8.31 \times 293.0} = [0.0789 \text{ mol}] \text{ (1)}$ $nN_A = 0.0789 \times 6.02 \times 10^{23} = 4.75 \times 10^{22} \text{ (1)}$ Alternative: $N = \frac{pV}{kT} = \frac{1.2 \times 10^5 \times 1.6 \times 10^{-3}}{1.38 \times 10^{-23} \times 293.0} \text{ (1)}$ $= 4.75 \times 10^{22} \text{ (1)}$		2		2	2	
		(ii)	Valid method e.g. $\frac{1}{2}mv^2 = \frac{3}{2}kT \text{ (1)}$ Answer = 1 350 [m s ⁻¹] ecf on $N \text{ (1)}$	1	1		2	2	
	(b)	(i)	Combine: $nRT = \frac{1}{3}Nm\overline{c^2} \text{ (1)}$ Substitution of $N = nN_A$ or $n = 1$ so $N = N_A \text{ (1)}$ KE [of a molecule] = $\frac{1}{2}m\overline{c^2}$ or $U = N\frac{1}{2}m\overline{c^2}$ can be implied if everything is clear (1) So use or implied $k = \frac{R}{N_A}$ and convincing algebra (1) Common alternative for full marks: $nRT = \frac{1}{3}Nm\overline{c^2} \text{ (1)}$ $NkT = \frac{1}{3}Nm\overline{c^2} \text{ (1)}$ (implies 1 st line if 1 st line omitted) $KE = \frac{1}{2}m\overline{c^2} \text{ (1)}$ (can be implied if everything clear) Algebra leading to $KE = \frac{3}{2}kT \text{ (1)}$	4			4	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			Another alternative for full marks: $KE = \frac{1}{2} m \overline{c^2}$ (1) Substituting from $pV = \frac{1}{3} Nm \overline{c^2}$ into above e.g. $KE = \frac{1}{2} \frac{3pV}{N}$ (1) Substituting $pV = NkT$ into above e.g. $KE = \frac{1}{2} \frac{3NkT}{N}$ (1) Hence $KE = \frac{3}{2} kT$ (1) Alternative for a maximum of 3 marks: Internal energy or $U = \frac{3}{2} nRT$ (1) $U = \frac{3}{2} NkT$ (1) This is for N molecules, hence $KE = \frac{3}{2} kT$ (1)						
		(ii)	$\frac{3}{2} \times 1.38 \times 10^{-23} \times 293 = 6.07 \times 10^{-21}$ [J]		1		1	1	
		(iii)	4.75×10^{22} ecf $\times 6.07 \times 10^{-21}$ ecf = 288 [J]		1		1	1	
	(c)	(i)	Ideal gas – kinetic energy only or no PE (1) Liquid – kinetic energy and potential energy (1) Link PE to intermolecular forces or separation of molecules or intermolecular force / separation mentioned for liquid (1)	3			3		
		(ii)	At absolute zero or 0 [K] or -273 [°C] Do not accept 0 °C OR -273 K (treat as neutral, don't penalise – they may have said something else that is correct e.g. absolute zero)	1			1		
			Question 2 total	9	5	0	14	8	0